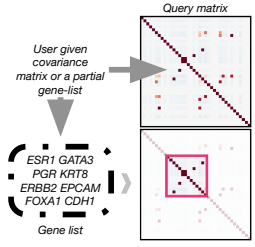
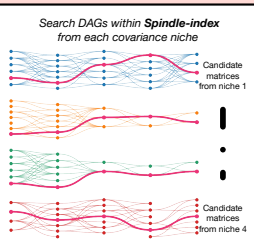


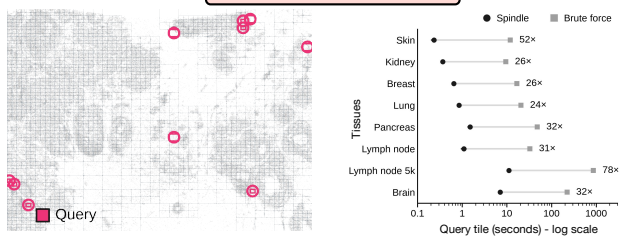
D Covariance queries for **Spindle**

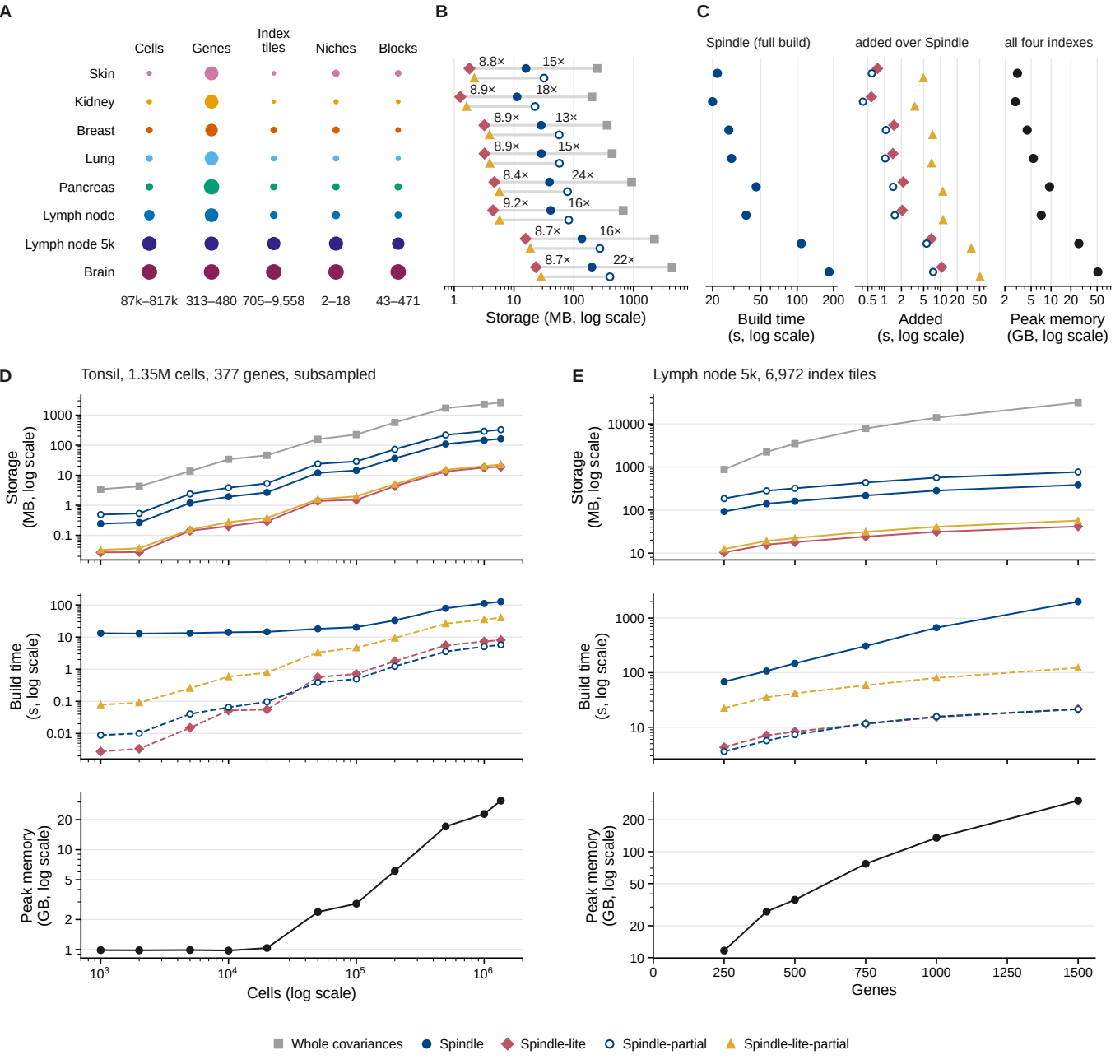


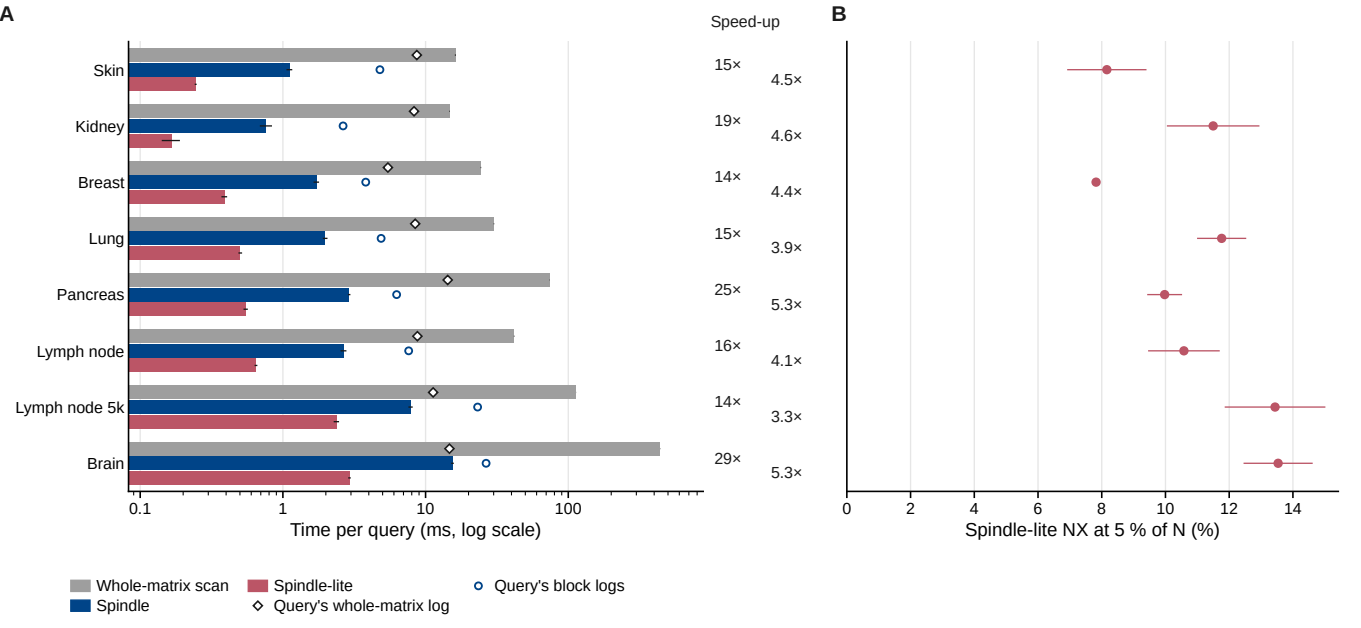
E DAG based search in **Spindle-index**

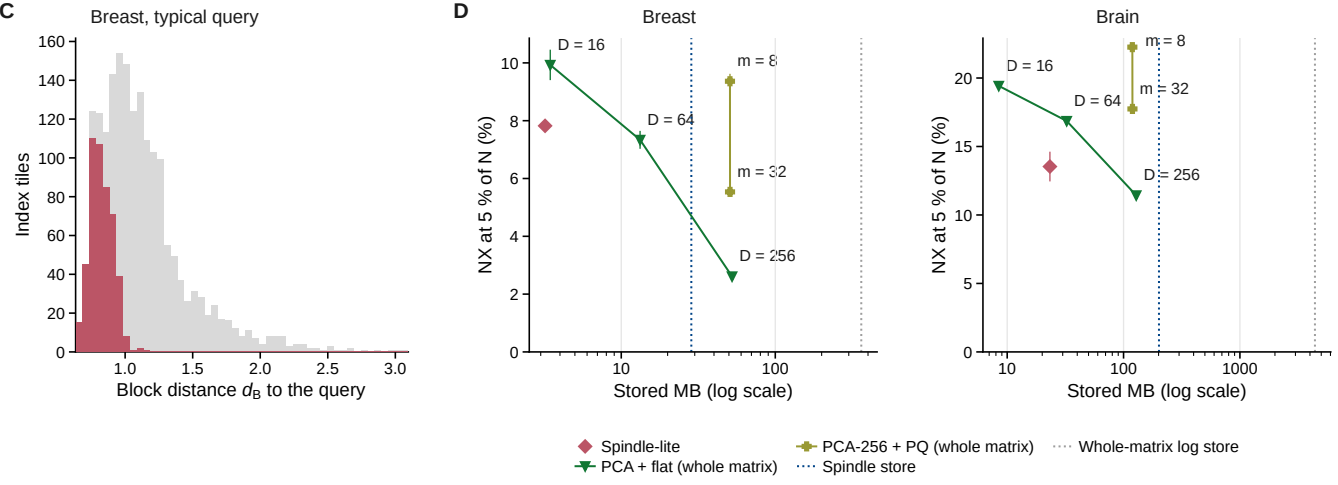


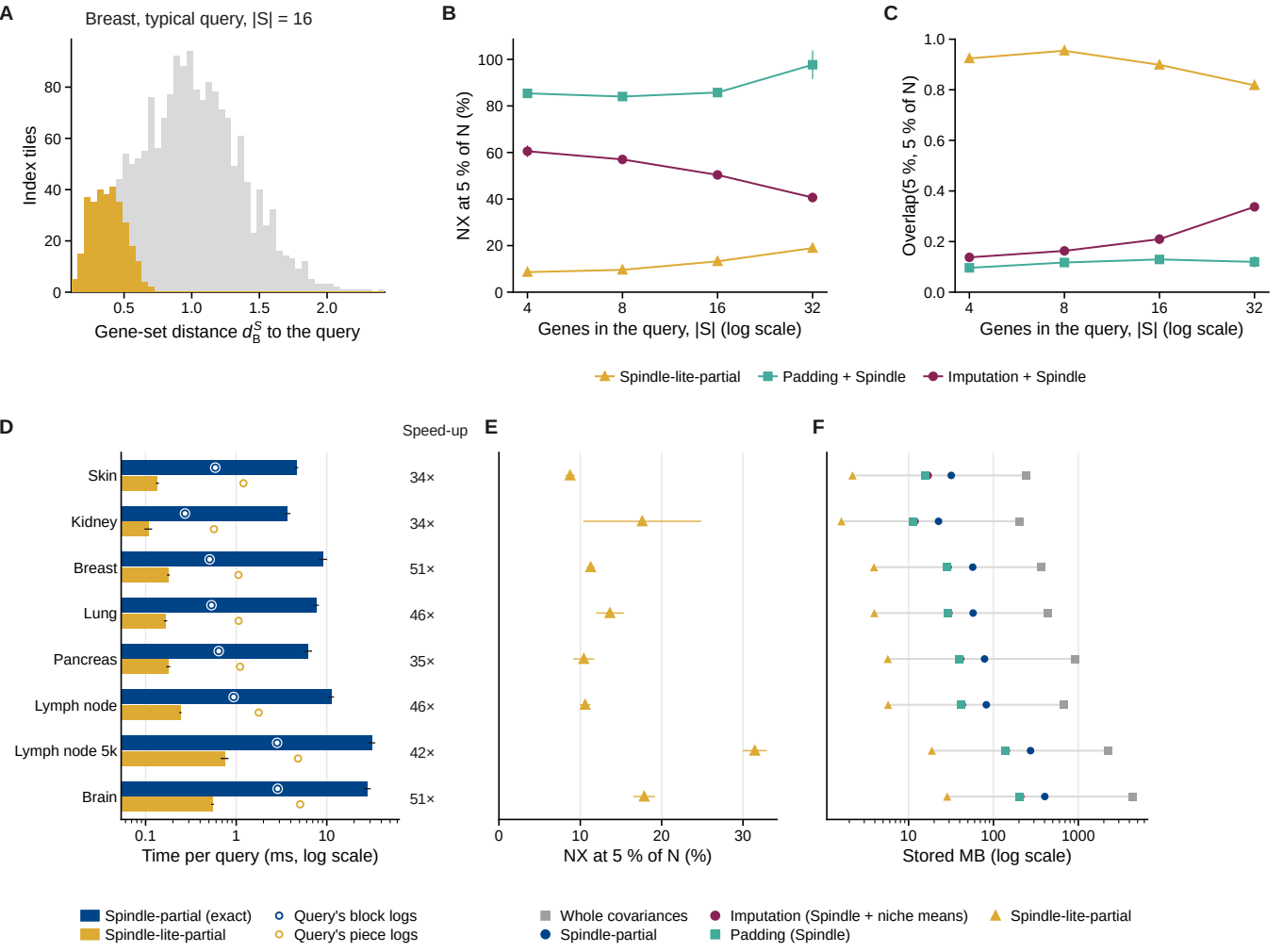
F Fast retrieval of query results

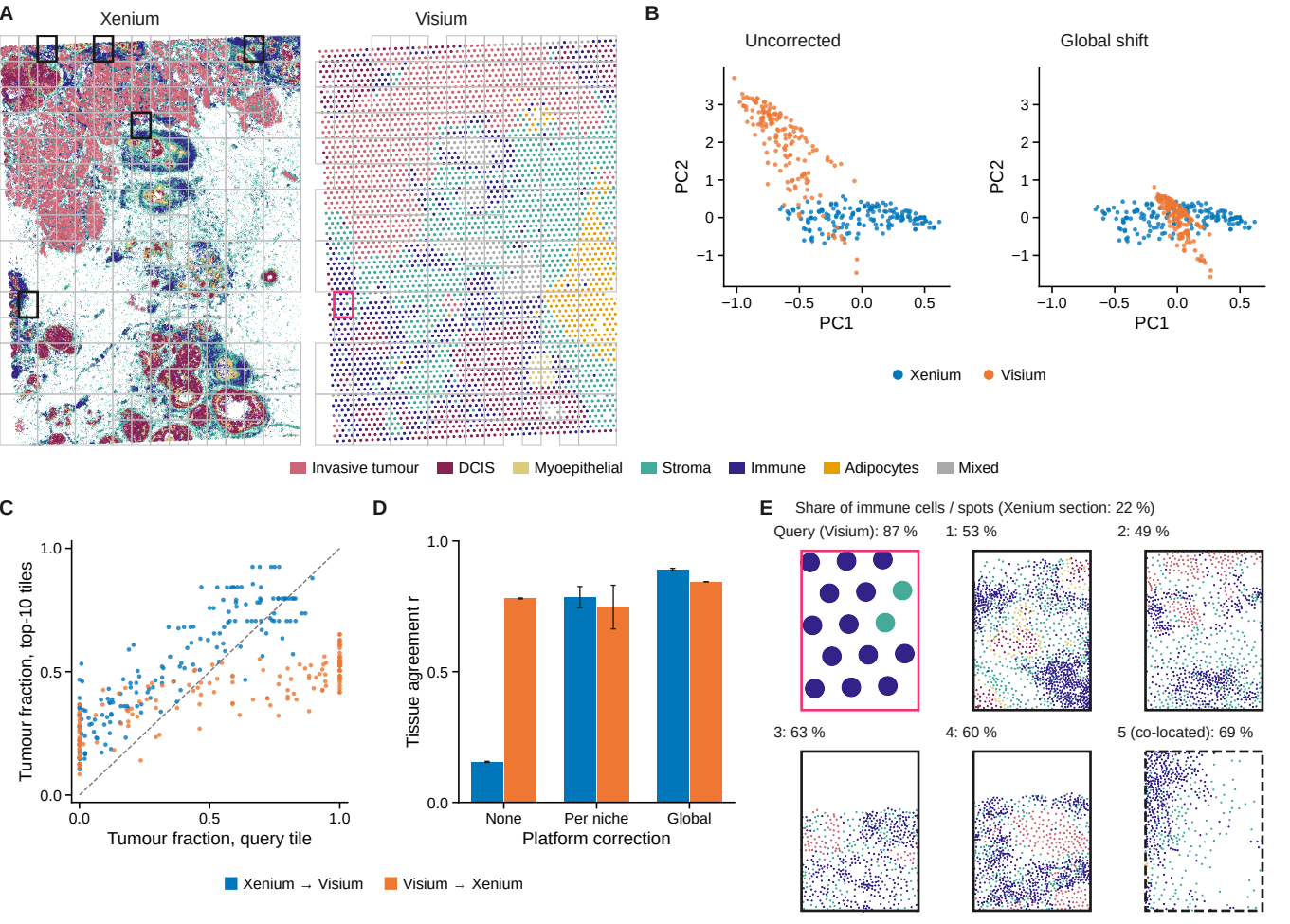


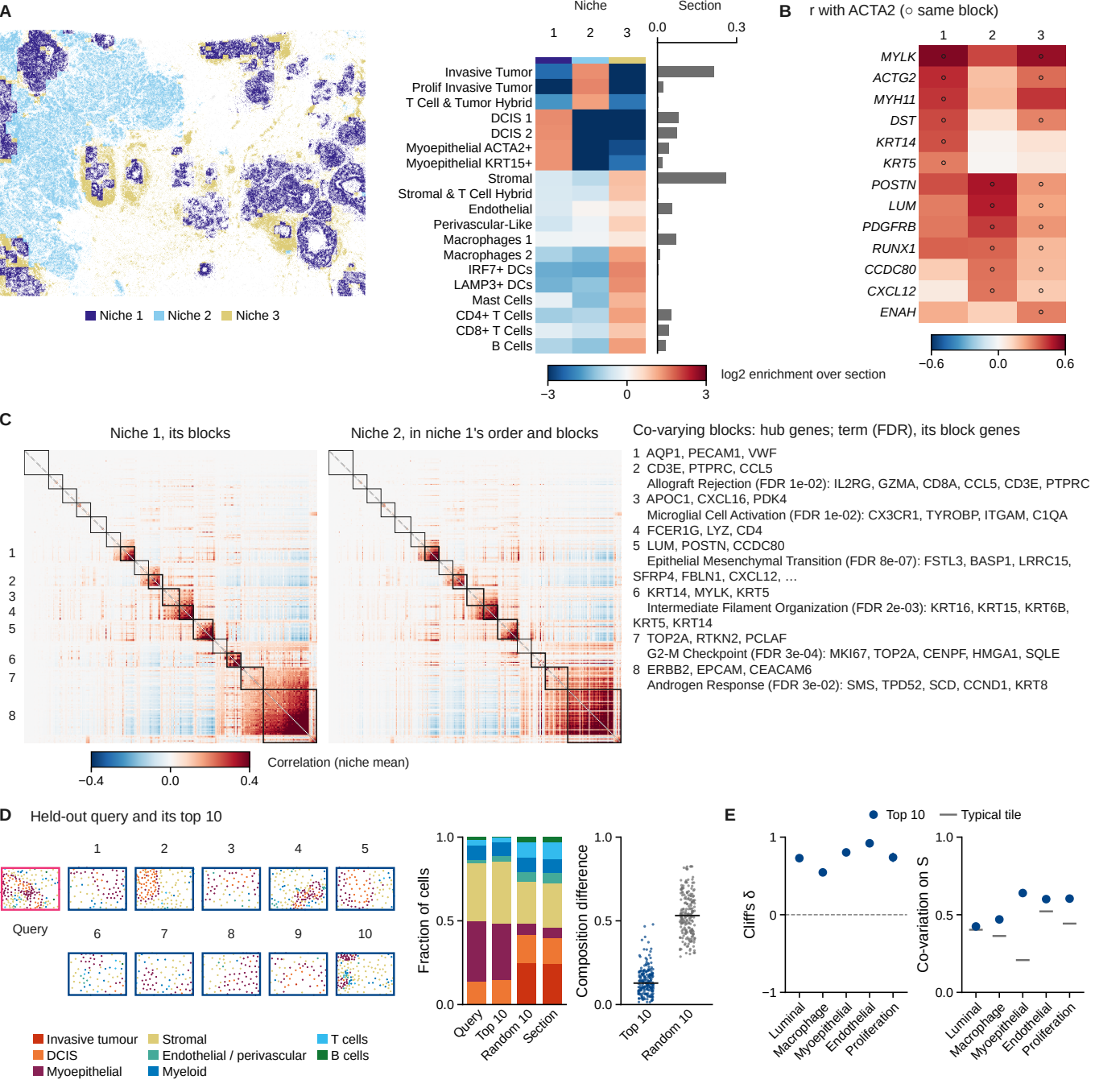

A
Cells
Genes
Index tiles
Niches
Blocks
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain
87k–817k
313–480
705–9,558
2–18
43–471
B
8.8×
15×
8.9×
18×
8.9×
13×
8.9×
15×
8.4×
24×
9.2×
16×
8.7×
16×
8.7×
22×
Storage (MB, log scale)
C
Spindle (full build)
added over Spindle
all four indexes
Build time (s, log scale)
Added (s, log scale)
Peak memory (GB, log scale)
D
Tonsil, 1.35M cells, 377 genes, subsampled
Storage (MB, log scale)
Build time (s, log scale)
Peak memory (GB, log scale)
Cells (log scale)
E
Lymph node 5k, 6,972 index tiles
Storage (MB, log scale)
Build time (s, log scale)
Peak memory (GB, log scale)
Genes
Whole covariances
Spindle
Spindle-lite
Spindle-partial
Spindle-lite-partial

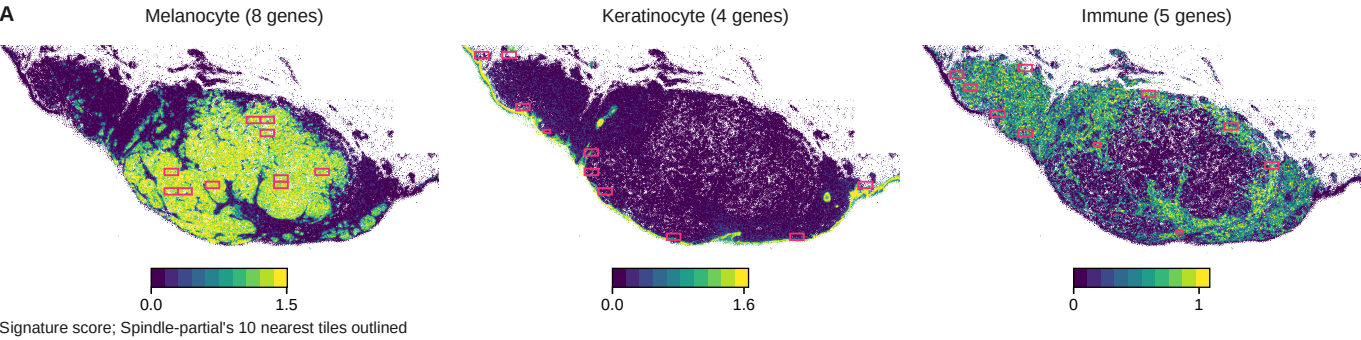

A
Speed-up
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain
15×
19×
14×
15×
25×
16×
14×
29×
4.5×
4.6×
4.4×
3.9×
5.3×
4.1×
3.3×
5.3×
0.1
1
10
100
Time per query (ms, log scale)
B
0
2
4
6
8
10
12
14
Spindle-lite NX at 5 % of N (%)
Whole-matrix scan
Spindle
Spindle-lite
Query's whole-matrix log
Query's block logs

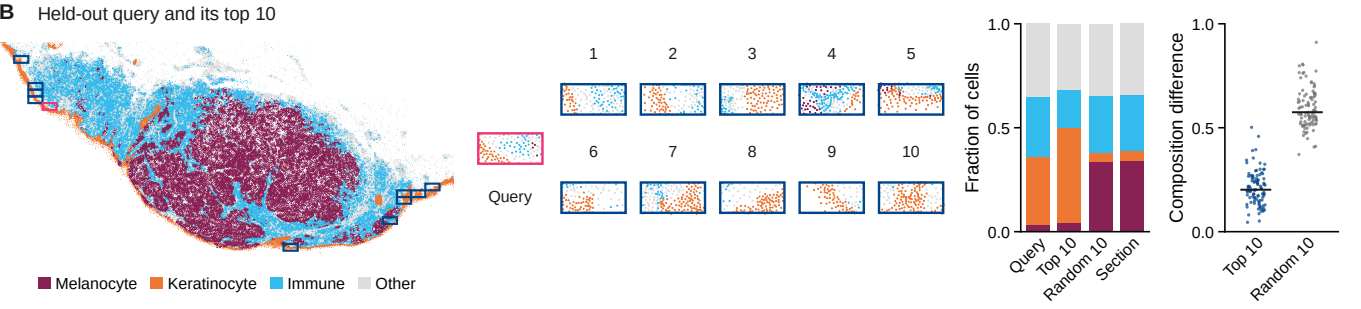

C
Breast, typical query
Index tiles
Block distance d_B to the query
D
Breast
Brain
NX at 5 % of N (%)
Stored MB (log scale)
D = 16
D = 64
D = 256
m = 8
m = 32
Spindle-lite
PCA + flat (whole matrix)
PCA-256 + PQ (whole matrix)
Spindle store
Whole-matrix log store

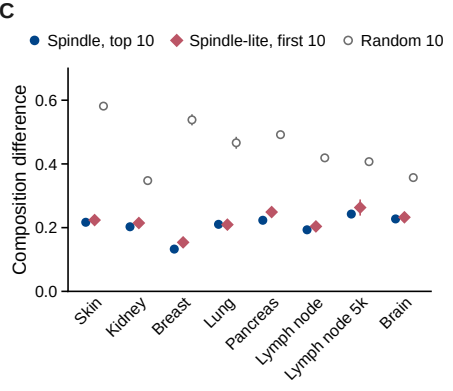

A
Breast, typical query, |S| = 16
Index tiles
Gene-set distance d_B^S to the query
B
NX at 5 % of N (%)
Genes in the query, |S| (log scale)
C
Overlap(5 %, 5 % of N)
Genes in the query, |S| (log scale)
Spindle-lite-partial
Padding + Spindle
Imputation + Spindle
D
Speed-up
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain
34×
34×
51×
46×
35×
46×
42×
51×
Time per query (ms, log scale)
E
NX at 5 % of N (%)
F
Stored MB (log scale)
Spindle-partial (exact)
Spindle-lite-partial
Query's block logs
Query's piece logs
Whole covariances
Spindle-partial
Imputation (Spindle + niche means)
Padding (Spindle)
Spindle-lite-partial

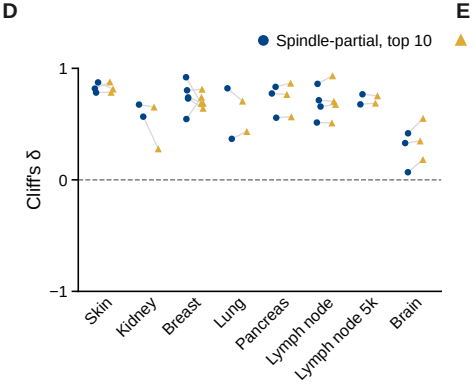

A
Xenium
Visium
B
Uncorrected
Global shift
PC2
PC1
Xenium
Visium
Invasive tumour
DCIS
Myoepithelial
Stroma
Immune
Adipocytes
Mixed
C
Tumour fraction, top-10 tiles
Tumour fraction, query tile
D
Tissue agreement r
None
Per niche
Global
Platform correction
Xenium → Visium
Visium → Xenium
E
Share of immune cells / spots (Xenium section: 22 %)
Query (Visium): 87 %
1: 53 %
2: 49 %
3: 63 %
4: 60 %
5 (co-located): 69 %

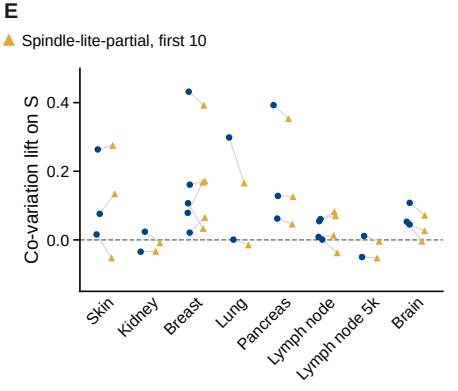

A
Niche 1
Niche 2
Niche 3
Niche
Section
1
2
3
0.0
0.3
Invasive Tumor
Prolif Invasive Tumor
T Cell & Tumor Hybrid
DCIS 1
DCIS 2
Myoepithelial ACTA2+
Myoepithelial KRT15+
Stromal
Stromal & T Cell Hybrid
Endothelial
Perivascular-Like
Macrophages 1
Macrophages 2
IRF7+ DCs
LAMP3+ DCs
Mast Cells
CD4+ T Cells
CD8+ T Cells
B Cells
−3
0
3
log2 enrichment over section
B
r with ACTA2 (○ same block)
1
2
3
MYLK
ACTG2
MYH11
DST
KRT14
KRT5
POSTN
LUM
PDGFRB
RUNX1
CCDC80
CXCL12
ENAH
−0.6
0.0
0.6
C
Niche 1, its blocks
Niche 2, in niche 1's order and blocks
1
2
3
4
5
6
7
8
−0.4
0.0
0.4
Correlation (niche mean)
Co-varying blocks: hub genes; term (FDR), its block genes
1 AQP1, PECAM1, VWF
2 CD3E, PTPRC, CCL5
Allograft Rejection (FDR 1e-02): IL2RG, GZMA, CD8A, CCL5, CD3E, PTPRC
3 APOC1, CXCL16, PDK4
Microglial Cell Activation (FDR 1e-02): CX3CR1, TYROBP, ITGAM, C1QA
4 FCER1G, LYZ, CD4
5 LUM, POSTN, CCDC80
Epithelial Mesenchymal Transition (FDR 8e-07): FSTL3, BASP1, LRRC15, SFRP4, FBLN1, CXCL12, …
6 KRT14, MYLK, KRT5
Intermediate Filament Organization (FDR 2e-03): KRT16, KRT15, KRT6B, KRT5, KRT14
7 TOP2A, RTKN2, PCLAF
G2-M Checkpoint (FDR 3e-04): MKI67, TOP2A, CENPF, HMGA1, SQLE
8 ERBB2, EPCAM, CEACAM6
Androgen Response (FDR 3e-02): SMS, TPD52, SCD, CCND1, KRT8
D
Held-out query and its top 10
1
2
3
4
5
Query
6
7
8
9
10
Invasive tumour
DCIS
Myoepithelial
Stromal
Endothelial / perivascular
Myeloid
T cells
B cells
Fraction of cells
1.0
0.5
0.0
Query
Top 10
Random 10
Section
Composition difference
Top 10
Random 10
E
Top 10
Typical tile
Cliff's δ
1
0
−1
Luminal
Macrophage
Myoepithelial
Endothelial
Proliferation
Co-variation on S
1.0
0.5
0.0

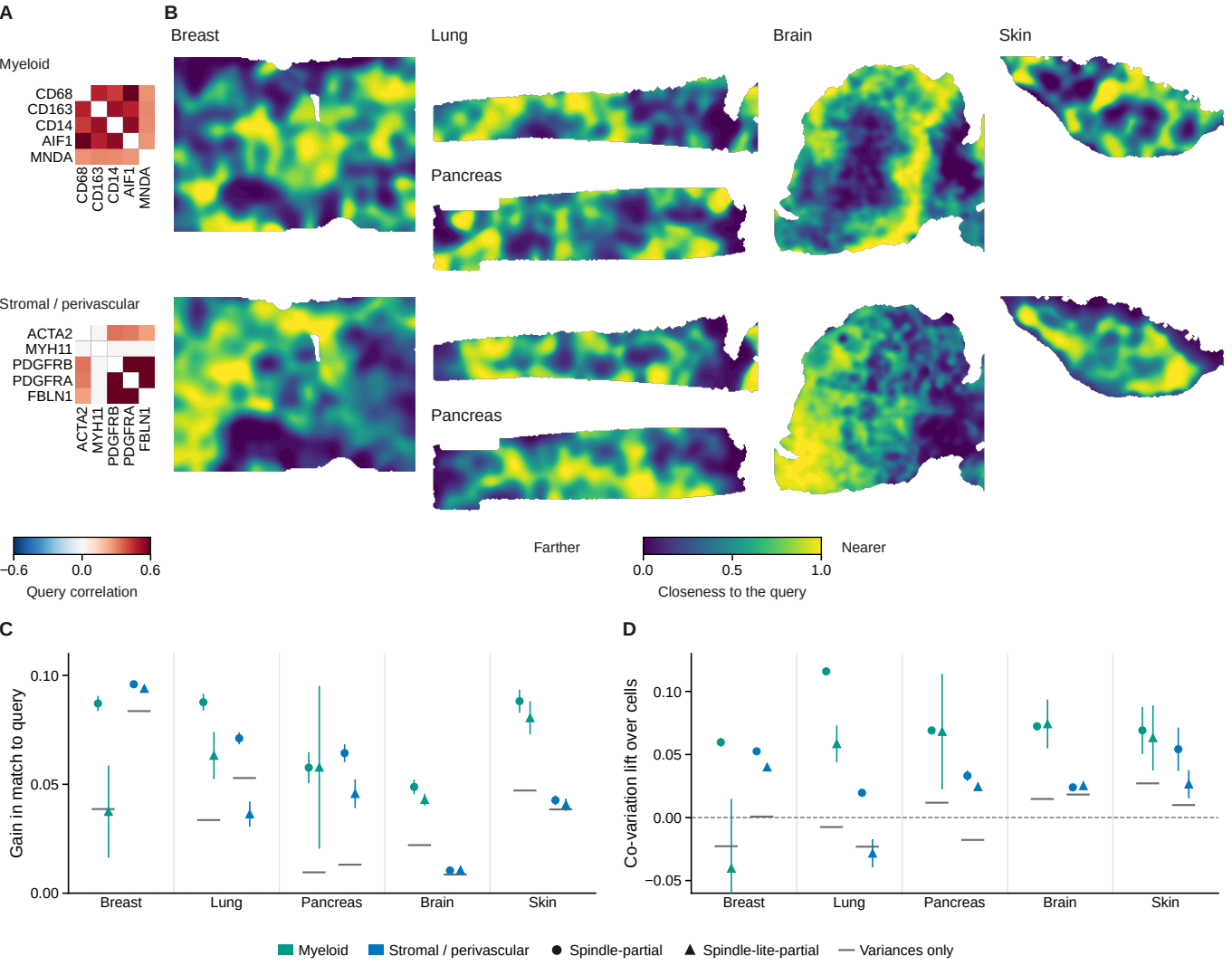

A
Melanocyte (8 genes)
Keratinocyte (4 genes)
Immune (5 genes)
0.0
1.5
0.0
1.6
0
1
Signature score; Spindle-partial's 10 nearest tiles outlined


B
Held-out query and its top 10
1
2
3
4
5
6
7
8
9
10
Query
Melanocyte
Keratinocyte
Immune
Other
Fraction of cells
1.0
0.5
0.0
Query
Top 10
Random 10
Section
Composition difference
1.0
0.5
0.0
Top 10
Random 10


C
Spindle, top 10
Spindle-lite, first 10
Random 10
Composition difference
0.6
0.4
0.2
0.0
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain


D
Spindle-partial, top 10
Cliff's δ
1
0
−1
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain


E
Spindle-lite-partial, first 10
Co-variation lift on S
0.4
0.2
0.0
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain


A
B
Breast
Lung
Brain
Skin
Myeloid
CD68
CD163
CD14
AIF1
MNDA
Pancreas
Stromal / perivascular
ACTA2
MYH11
PDGFRB
PDGFRA
FBLN1
−0.6
0.0
0.6
Query correlation
Farther
Nearer
0.0
0.5
1.0
Closeness to the query

C
Gain in match to query
0.10
0.05
0.00
Breast
Lung
Pancreas
Brain
Skin
D
Co-variation lift over cells
0.10
0.05
0.00
−0.05
Breast
Lung
Pancreas
Brain
Skin
Myeloid
Stromal / perivascular
Spindle-partial
Spindle-lite-partial
Variances only